

Assignment

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# **Analysing Company Performance with SQL**

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## 1. Executive Summary

**Project Overview:** This project aims to provide business insights based on data through SQL analysis across key commercial areas.

**Problem statement:** Various departments require analytical support for decision-making.

**Achieved outcomes and results:** Successfully generated detailed reports to support multiple departments, including pricing strategy, shipping assessment, employee management and sales KPI.



## 2. Introduction

### **The key objectives of the project:**

- Analyze the pricing strategy of products and evaluate competitive positioning across several categories.
- Assess the supplier inventory status and shipping performance in 1998.
- Generate a detailed report on the structure of employees, from basic staff to management levels, including employee demographics and key hiring information.
- Develop KPI frameworks to measure employee sales performance.

### **The stakeholders' requirements:**

Product Team: A specific products list, with pricing range from \$20 to \$50 and not discontinued.

Logistics Team: Recapitulation of shipping situation in 1998, indicating which countries' shipping date over 5 dates and total number is greater 10 orders; In addition, point out which month's order is over 35 totally in 1997.

HR Team: a detailed report on the structure of employees, from basic staff to management levels, including employee demographics and key hiring information.

Pricing Team: summary those products, with increasing not between 20% and 30%. In terms of category, finding each product's price position in belonging's category, and the total amount and volume orders in each category, and each products' position compared to category average and median unit price.

Sales Team: a completed KPI to get through sales amount, unique and not unique number orders totally, discount and sales amount totally, for each employee.



### 3. Dataset presentation

- Acquire the dataset from:  
[https://drive.google.com/file/d/1xfUF79LxsYte9Wevce-Fhpw50uiS\\_0Jx/view](https://drive.google.com/file/d/1xfUF79LxsYte9Wevce-Fhpw50uiS_0Jx/view)
- The data sources coming from a company
- Columns introduce
  - unit\_price: to select specific price products
  - order\_id: to identify each order
  - employee\_id: to distribute sales performance



## 4. Business Question 1

### a. Description

Base on the annual review of pricing strategy, the Product Team products requested a filtered list of active products priced between 20 and 50, sorted by unit price in descending order.

### b. Results

There are 27 products totally meeting the requirements above, including specific pricing range from 20 to 50 and not discontinued.

|    | AZ product_name                  | unit_price |
|----|----------------------------------|------------|
| 1  | Chef Anton's Cajun Seasoning     | 22         |
| 2  | Grandma's Boysenberry Spread     | 25         |
| 3  | Uncle Bob's Organic Dried Pears  | 30         |
| 4  | Northwoods Cranberry Sauce       | 40         |
| 5  | Ikura                            | 31         |
| 6  | Queso Cabrales                   | 21         |
| 7  | Queso Manchego La Pastora        | 38         |
| 8  | Tofu                             | 23.25      |
| 9  | Gustaf's Knäckebröd              | 21         |
| 10 | Gumbär Gummibärchen              | 31.23      |
| 11 | Schoggi Schokolade               | 43.9       |
| 12 | Nord-Ost Matjeshering            | 25.89      |
| 13 | Mascarpone Fabioli               | 32         |
| 14 | Gravad lax                       | 26         |
| 15 | Ipoh Coffee                      | 46         |
| 16 | Maxilaku                         | 20         |
| 17 | Pâté chinois                     | 24         |
| 18 | Gnocchi di nonna Alice           | 38         |
| 19 | Camembert Pierrot                | 34         |
| 20 | Sirop d'érable                   | 28.5       |
| 21 | Tarte au sucre                   | 49.3       |
| 22 | Vegie-spread                     | 43.9       |
| 23 | Wimmers gute Semmelknödel        | 33.25      |
| 24 | Louisiana Fiery Hot Pepper Sauce | 21.05      |
| 25 | Gudbrandsdalsost                 | 36         |
| 26 | Flotemysost                      | 21.5       |
| 27 | Mozzarella di Giovanni           | 34.8       |

### c. Key insights and Findings

It reveals that this company has 27 products continued and achieved the requirements. This finding confirms that the company offered diverse and active products within target price range.



## 5. Business Question 2

### a. Description

Due to examine the performance of countries and find which one had the bad-performance, give the Logistics Team a countries lists about the average days between the shipping date and the order date and the total unique order numbers based on each order id. Meanwhile, only the order date is 1998, average date is greater or equal 5 days and the total orders is more than 10 orders are showed.

### b. Results

By these limitations above, there are 9 countries identified as bad-performance.

### c. Key insights and Findings

When making strategy to improve, the company should priority take these 9 countries into consideration.

|   | 123 | AZ country ▼ | 123 total_number ▼ | 123 average ▼ |
|---|-----|--------------|--------------------|---------------|
| 1 |     | Austria      | 11                 | 5.89          |
| 2 |     | Brazil       | 28                 | 8.12          |
| 3 |     | France       | 23                 | 9.43          |
| 4 |     | Germany      | 34                 | 5.38          |
| 5 |     | Spain        | 12                 | 7.83          |
| 6 |     | Sweden       | 14                 | 13.29         |
| 7 |     | UK           | 16                 | 6.25          |
| 8 |     | USA          | 39                 | 7.89          |
| 9 |     | Venezuela    | 18                 | 8.73          |



## 6. Business Question 3

### a. Description

This analysis examines the organizational structure for the HR Team, including each employee's full name, job title and age at hiring. Additionally, this report provides their manager's full name and job title, which enables the HR Team to examine the organizational structure.

### b. Results

According to this data, there are nine employees in this company. Six of nine hold the title of Sales Representative, followed by inside sales coordinator, sales manager and Vice President Sales.

Five of nine are managed by the Vice President of Sales, while the Sales Manager oversees three employees. The only person who does not report to anyone is the Vice President of Sales.

In terms of age, six of nine are under 40 years old, and other remaining employees are over 40 years old.

|   | AZ employees_full_name | AZ title                 | 123 age | AZ managers_full_name | AZ managers_title     |
|---|------------------------|--------------------------|---------|-----------------------|-----------------------|
| 1 | Anne Dodsworth         | Sales Representative     | 28      | Steven Buchanan       | Sales Manager         |
| 2 | Janet Leverling        | Sales Representative     | 28      | Andrew Fuller         | Vice President, Sales |
| 3 | Michael Suyama         | Sales Representative     | 30      | Steven Buchanan       | Sales Manager         |
| 4 | Robert King            | Sales Representative     | 33      | Steven Buchanan       | Sales Manager         |
| 5 | Laura Callahan         | Inside Sales Coordinator | 36      | Andrew Fuller         | Vice President, Sales |
| 6 | Steven Buchanan        | Sales Manager            | 38      | Andrew Fuller         | Vice President, Sales |
| 7 | Andrew Fuller          | Vice President, Sales    | 40      | [NULL]                | [NULL]                |
| 8 | Nancy Davolio          | Sales Representative     | 43      | Andrew Fuller         | Vice President, Sales |
| 9 | Margaret Peacock       | Sales Representative     | 55      | Andrew Fuller         | Vice President, Sales |

### c. Key insights and Findings

Given the outcome above, we can see that the company's reporting structure is that five of nine are managed by the Vice President of Sales, and three of nine are managed by the Sales Manager.

This indicates that the company might benefit from implementing a more comprehensive hierarchical structure.

This company consists of mature employees (almost over 30) pointing that it is an experienced firm. To maintain vitality, they should consider hiring younger employees.



## 7. Business Question 4

### a. Description

This query from the Logistics Team investigates the overview of performance around the world from 1997-1998 to discover the month with outstanding performance. This includes each month's information, total number of orders and freight. Moreover, the order date should be from 1997 to 1998, and the total order count should exceed 35.

### b. Results

From this table, it is clear that April 1998 recorded the highest total freight among all the listed months, at 6394, with the highest number of orders at 74, followed by January 1998 and March 1998, at 5463 and 5379 respectively. However, September 1997 showed the lowest values in both freight and orders, at 3,237 and 37 respectively.

|   | AZ the_date ▼ | 123 total_number_of_orders ▼ | 123 total_freight ▼ |
|---|---------------|------------------------------|---------------------|
| 1 | 1998-04-01    | 74                           | 6,394               |
| 2 | 1998-01-01    | 55                           | 5,463               |
| 3 | 1998-03-01    | 73                           | 5,379               |
| 4 | 1998-02-01    | 54                           | 4,273               |
| 5 | 1997-10-01    | 38                           | 3,946               |
| 6 | 1997-12-01    | 48                           | 3,758               |
| 7 | 1997-09-01    | 37                           | 3,237               |

### c. Key insights and Findings

This list points out that April 1998 has the best performance, and January to March show excellent numbers, which indicates that the company should focus their advertising on the first half of the year.



## 8. Business Question 5

### a. Description

Conducting an analysis of pricing strategy for the Pricing Team provides a product list showing each product's name, current and initial unit price, and the increasing percentage. It is required to filter the items by excluding percentage increases between 20% and 30%.

### b. Results

There are only two products meeting the requirements, one is Singaporean Hokkien Fried Mee, which had a 43% increase, with an initial price of 9.80 and a current price of 14.00. Another product, Queso Cabrales, had the highest increase percentage at 50%, with an initial price of 14.00 and a current price of 21.00.

|   | AZ product_name               | AZ initial | AZ current | 123 percentage |
|---|-------------------------------|------------|------------|----------------|
| 1 | Singaporean Hokkien Fried Mee | 9.80       | 14.00      | 43             |
| 2 | Queso Cabrales                | 14.00      | 21.00      | 50             |

### c. Key insights and Findings

This data presents two products with high percentage increases among a total of 77 products, giving the company insight into how these two products were impacted by the significant price increase.



## 9. Business Question 6

### a. Description

The query from the Pricing Team assesses each product category based on the price range defined by the team, including “1. Below \$20”, “2. \$20 - \$50” and “3. Over \$50”, along with each category’s name and price range, the total sales amount after discount and the volume of orders.

### b. Results

According to this table, eight product categories are listed with three pricing ranges, a total of twenty-four combinations, showing the total amount and the number of orders for each combination.

|    | A-Z category_name | A-Z price_range | 123 amount | 123 number_of_order |
|----|-------------------|-----------------|------------|---------------------|
| 1  | Beverages         | 1. Below \$20   | 102,945    | 317                 |
| 2  | Beverages         | 2. \$20 - \$50  | 23,527     | 28                  |
| 3  | Beverages         | 3.Over \$50     | 141,397    | 24                  |
| 4  | Condiments        | 1. Below \$20   | 17,383     | 62                  |
| 5  | Condiments        | 2. \$20 - \$50  | 78,748     | 121                 |
| 6  | Condiments        | 3.Over \$50     | 9,916      | 24                  |
| 7  | Confections       | 1. Below \$20   | 53,365     | 197                 |
| 8  | Confections       | 2. \$20 - \$50  | 91,429     | 106                 |
| 9  | Confections       | 3.Over \$50     | 22,563     | 16                  |
| 10 | Dairy Products    | 1. Below \$20   | 16,569     | 81                  |
| 11 | Dairy Products    | 2. \$20 - \$50  | 146,783    | 204                 |
| 12 | Dairy Products    | 3.Over \$50     | 71,156     | 54                  |
| 13 | Grains/Cereals    | 1. Below \$20   | 16,410     | 76                  |
| 14 | Grains/Cereals    | 2. \$20 - \$50  | 71,673     | 91                  |
| 15 | Grains/Cereals    | 3.Over \$50     | 7,662      | 23                  |
| 16 | Meat/Poultry      | 1. Below \$20   | 4,728      | 36                  |
| 17 | Meat/Poultry      | 2. \$20 - \$50  | 70,699     | 96                  |
| 18 | Meat/Poultry      | 3.Over \$50     | 87,595     | 36                  |
| 19 | Produce           | 1. Below \$20   | 2,432      | 13                  |
| 20 | Produce           | 2. \$20 - \$50  | 55,732     | 81                  |
| 21 | Produce           | 3.Over \$50     | 41,820     | 39                  |
| 22 | Seafood           | 1. Below \$20   | 65,110     | 217                 |
| 23 | Seafood           | 2. \$20 - \$50  | 36,980     | 70                  |
| 24 | Seafood           | 3.Over \$50     | 29,172     | 27                  |



### c. Key insights and Findings

The table shows the pricing structure of each category, including total amount and number of orders, which successfully provides the company with a broad overview to review its pricing strategy and extract relevant insights.



## 10. Business Question 7

### a. Description

This report provides an overview of the current inventory status across regions - America, Europe, Asia-Pacific- covering each corresponding category name, total stock and units on order, and reorder levels.

### b. Results

This outcome shows there are eight product categories in total, most of which are distributed across three regions, except Dairy Products which appear only in one region, and Meat/Poultry in two. The highest stock level is found in Seafood in Europe (396 units), followed by Dairy Products in Europe (393 units), with Meat/Poultry in Europe showing zero across all columns.

|    | A-Z category_name | A-Z area     | 123 total_in_stock | 123 total_in_order | 123 total_in_reorder |
|----|-------------------|--------------|--------------------|--------------------|----------------------|
| 1  | Beverages         | America      | 203                | 0                  | 40                   |
| 2  | Beverages         | Asia-Pacific | 32                 | 20                 | 55                   |
| 3  | Beverages         | Europe       | 324                | 40                 | 100                  |
| 4  | Condiments        | America      | 372                | 100                | 70                   |
| 5  | Condiments        | Asia-Pacific | 90                 | 0                  | 25                   |
| 6  | Condiments        | Europe       | 45                 | 70                 | 40                   |
| 7  | Confections       | America      | 17                 | 0                  | 0                    |
| 8  | Confections       | Asia-Pacific | 29                 | 0                  | 10                   |
| 9  | Confections       | Europe       | 340                | 180                | 155                  |
| 10 | Dairy Products    | Europe       | 393                | 140                | 110                  |
| 11 | Grains/Cereals    | Asia-Pacific | 64                 | 0                  | 25                   |
| 12 | Grains/Cereals    | Europe       | 244                | 90                 | 130                  |
| 13 | Meat/Poultry      | America      | 136                | 0                  | 30                   |
| 14 | Meat/Poultry      | Asia-Pacific | 29                 | 0                  | 0                    |
| 15 | Meat/Poultry      | Europe       | 0                  | 0                  | 0                    |
| 16 | Produce           | America      | 15                 | 0                  | 10                   |
| 17 | Produce           | Asia-Pacific | 59                 | 20                 | 15                   |
| 18 | Produce           | Europe       | 26                 | 0                  | 0                    |
| 19 | Seafood           | America      | 208                | 0                  | 40                   |
| 20 | Seafood           | Asia-Pacific | 97                 | 0                  | 5                    |
| 21 | Seafood           | Europe       | 396                | 120                | 100                  |

### c. Key insights and Findings

The detailed information enables a clear review of inventory distribution by category and region, including these metrics. This highlights the relationship between stock, order and reorder levels, enabling the company to validate supply sufficiency and identify areas that may require restocking.



## 11. Business Question 8

### a. Description

This request from the Pricing Team investigates the current products pricing by calculating the average and median unit price for each category, then comparing each product's unit price with its category's average and median, with the outcome classified as "Below", "Equal" and "Over".

### b. Results

This table shows each product's position within its category, including the product's name, its price, and how the price compares to the average and median price in the same category. There are forty-three products classified as Below Average, while the remaining items fall into the Over Average group. The distribution between Below Median Over Median is evenly split.



|    | AZ category_name | AZ product_name                  | 123 unit_price | 123 avg_price | 123 median_price | AZ position_against_c_avg | AZ position_against_median |
|----|------------------|----------------------------------|----------------|---------------|------------------|---------------------------|----------------------------|
| 1  | Beverages        | Chartreuse verte                 | 18             | 46.03         | 18               | Below Average             | Equal Median               |
| 2  | Beverages        | Côte de Blaye                    | 263.5          | 46.03         | 18               | Over Average              | Over Median                |
| 3  | Beverages        | Iphoh Coffee                     | 46             | 46.03         | 18               | Below Average             | Over Median                |
| 4  | Beverages        | Lakkalikööri                     | 18             | 46.03         | 18               | Below Average             | Equal Median               |
| 5  | Beverages        | Laughing Lumberjack Lager        | 14             | 46.03         | 18               | Below Average             | Below Median               |
| 6  | Beverages        | Outback Lager                    | 15             | 46.03         | 18               | Below Average             | Below Median               |
| 7  | Beverages        | Rhönbräu Klosterbier             | 7.75           | 46.03         | 18               | Below Average             | Below Median               |
| 8  | Beverages        | Sasquatch Ale                    | 14             | 46.03         | 18               | Below Average             | Below Median               |
| 9  | Beverages        | Steeleye Stout                   | 18             | 46.03         | 18               | Below Average             | Equal Median               |
| 10 | Condiments       | Aniseed Syrup                    | 10             | 22.99         | 21.05            | Below Average             | Below Median               |
| 11 | Condiments       | Chef Anton's Cajun Seasoning     | 22             | 22.99         | 21.05            | Below Average             | Over Median                |
| 12 | Condiments       | Genen Shouyu                     | 13             | 22.99         | 21.05            | Below Average             | Below Median               |
| 13 | Condiments       | Grandma's Boysenberry Spread     | 25             | 22.99         | 21.05            | Over Average              | Over Median                |
| 14 | Condiments       | Gula Malacca                     | 19.45          | 22.99         | 21.05            | Below Average             | Below Median               |
| 15 | Condiments       | Louisiana Fiery Hot Pepper Sauce | 21.05          | 22.99         | 21.05            | Below Average             | Equal Median               |
| 16 | Condiments       | Louisiana Hot Spiced Okra        | 17             | 22.99         | 21.05            | Below Average             | Below Median               |
| 17 | Condiments       | Northwoods Cranberry Sauce       | 40             | 22.99         | 21.05            | Over Average              | Over Median                |
| 18 | Condiments       | Original Frankfurter grüne Soße  | 13             | 22.99         | 21.05            | Below Average             | Below Median               |
| 19 | Condiments       | Sirop d'érable                   | 28.5           | 22.99         | 21.05            | Over Average              | Over Median                |
| 20 | Condiments       | Veggie-spread                    | 43.9           | 22.99         | 21.05            | Over Average              | Over Median                |
| 21 | Confections      | Chocolade                        | 12.75          | 25.16         | 16.25            | Below Average             | Below Median               |
| 22 | Confections      | Gumbär Gummibärchen              | 31.23          | 25.16         | 16.25            | Over Average              | Over Median                |
| 23 | Confections      | Maxilaku                         | 20             | 25.16         | 16.25            | Below Average             | Over Median                |
| 24 | Confections      | NuNuCa Nuß-Nougat-Creme          | 14             | 25.16         | 16.25            | Below Average             | Below Median               |
| 25 | Confections      | Pavlova                          | 17.45          | 25.16         | 16.25            | Below Average             | Over Median                |
| 26 | Confections      | Schoggi Schokolade               | 43.9           | 25.16         | 16.25            | Over Average              | Over Median                |
| 27 | Confections      | Scottish Longbreads              | 12.5           | 25.16         | 16.25            | Below Average             | Below Median               |
| 28 | Confections      | Sir Rodney's Marmalade           | 81             | 25.16         | 16.25            | Over Average              | Over Median                |
| 29 | Confections      | Sir Rodney's Scones              | 10             | 25.16         | 16.25            | Below Average             | Below Median               |
| 30 | Confections      | Tarte au sucre                   | 49.3           | 25.16         | 16.25            | Over Average              | Over Median                |
| 31 | Confections      | Teatime Chocolate Biscuits       | 9.2            | 25.16         | 16.25            | Below Average             | Below Median               |
| 32 | Confections      | Valkoinen suklaa                 | 16.25          | 25.16         | 16.25            | Below Average             | Equal Median               |
| 33 | Confections      | Zaanse koeken                    | 9.5            | 25.16         | 16.25            | Below Average             | Below Median               |
| 34 | Dairy Products   | Camembert Pierrot                | 34             | 28.73         | 32               | Over Average              | Over Median                |
| 35 | Dairy Products   | Flotemysost                      | 21.5           | 28.73         | 32               | Below Average             | Below Median               |
| 36 | Dairy Products   | Geitost                          | 2.5            | 28.73         | 32               | Below Average             | Below Median               |
| 37 | Dairy Products   | Gorgonzola Telino                | 12.5           | 28.73         | 32               | Below Average             | Below Median               |
| 38 | Dairy Products   | Gudbrandsdalsost                 | 36             | 28.73         | 32               | Over Average              | Over Median                |
| 39 | Dairy Products   | Mascarpone Fabioli               | 32             | 28.73         | 32               | Over Average              | Equal Median               |
| 40 | Dairy Products   | Mozzarella di Giovanni           | 34.8           | 28.73         | 32               | Over Average              | Over Median                |
| 41 | Dairy Products   | Queso Cabrales                   | 21             | 28.73         | 32               | Below Average             | Below Median               |
| 42 | Dairy Products   | Queso Manchego La Pastora        | 38             | 28.73         | 32               | Over Average              | Over Median                |
| 43 | Dairy Products   | Raclette Courdavault             | 55             | 28.73         | 32               | Over Average              | Over Median                |
| 44 | Grains/Cereals   | Filo Mix                         | 7              | 21.29         | 19.5             | Below Average             | Below Median               |
| 45 | Grains/Cereals   | Gnocchi di nonna Alice           | 38             | 21.29         | 19.5             | Over Average              | Over Median                |
| 46 | Grains/Cereals   | Gustaf's Knäckebröd              | 21             | 21.29         | 19.5             | Below Average             | Over Median                |
| 47 | Grains/Cereals   | Ravioli Angelo                   | 19.5           | 21.29         | 19.5             | Below Average             | Equal Median               |
| 48 | Grains/Cereals   | Tunnbröd                         | 9              | 21.29         | 19.5             | Below Average             | Below Median               |
| 49 | Grains/Cereals   | Wimmers gute Semmelknödel        | 33.25          | 21.29         | 19.5             | Over Average              | Over Median                |
| 50 | Meat/Poultry     | Pâté chinois                     | 24             | 15.73         | 7.45             | Over Average              | Over Median                |
| 51 | Meat/Poultry     | Tourtière                        | 7.45           | 15.73         | 7.45             | Below Average             | Equal Median               |
| 52 | Produce          | Longlife Tofu                    | 10             | 29.06         | 23.25            | Below Average             | Below Median               |
| 53 | Produce          | Manjimup Dried Apples            | 53             | 29.06         | 23.25            | Over Average              | Over Median                |
| 54 | Produce          | Tofu                             | 23.25          | 29.06         | 23.25            | Below Average             | Equal Median               |
| 55 | Produce          | Uncle Bob's Organic Dried Pears  | 30             | 29.06         | 23.25            | Over Average              | Over Median                |
| 56 | Seafood          | Boston Crab Meat                 | 18.4           | 20.68         | 15               | Below Average             | Over Median                |
| 57 | Seafood          | Carnarvon Tigers                 | 62.5           | 20.68         | 15               | Over Average              | Over Median                |
| 58 | Seafood          | Escargots de Bourgogne           | 13.25          | 20.68         | 15               | Below Average             | Below Median               |
| 59 | Seafood          | Gravad lax                       | 26             | 20.68         | 15               | Over Average              | Over Median                |
| 60 | Seafood          | Ikura                            | 31             | 20.68         | 15               | Over Average              | Over Median                |
| 61 | Seafood          | Inlagd Sill                      | 19             | 20.68         | 15               | Below Average             | Over Median                |
| 62 | Seafood          | Jack's New England Clam Chowde   | 9.65           | 20.68         | 15               | Below Average             | Below Median               |
| 63 | Seafood          | Konbu                            | 6              | 20.68         | 15               | Below Average             | Below Median               |
| 64 | Seafood          | Nord-Ost Matjeshering            | 25.89          | 20.68         | 15               | Over Average              | Over Median                |
| 65 | Seafood          | Röd Kaviar                       | 15             | 20.68         | 15               | Below Average             | Equal Median               |
| 66 | Seafood          | Rogede sild                      | 9.5            | 20.68         | 15               | Below Average             | Below Median               |
| 67 | Seafood          | Spegesild                        | 12             | 20.68         | 15               | Below Average             | Below Median               |

### c. Key insights and Findings

According to this outcome, most product prices fall below the category average, particularly in the Beverages category, where eight out of nine products are classified as “Below Average”. This gives



the company insights into how comprehensive the price strategy is and enables informed decision making.



## 12. Business Question 9

### a. Description

This report, prepared for the Sales Team, provides a detailed analysis of each employee's sales performance, including their total sales amount both with and without discount, their total number of unique and overall orders. In addition, it also includes calculations of their average product amount and average order amount, as well as the total discount amount.

### b. Results

The highest total sales amount was recorded by Margaret Peacock, reaching \$250,187.45, followed by Nancy Davolio and Andrew Fuller, with 202,143.71 and 177,749.26 respectively. However, Anne Dodsworth performed the lowest sales amount at \$82,964.00.

|   | AZ full_name     | AZ title                 | 123 total_sales_excl_discount | 123 total_order_unique | 123 total_order | 123 avg_product_amount | 123 avg_order_amount | 123 total_discount_amount | 123 sales_amount_discount | 123 total_discount_percentage |
|---|------------------|--------------------------|-------------------------------|------------------------|-----------------|------------------------|----------------------|---------------------------|---------------------------|-------------------------------|
| 1 | Margaret Peacock | Sales Representative     | 250,187.45                    | 156                    | 420             | 595.68                 | 1,603.77             | 17,296.6                  | 232,890.85                | 6.91                          |
| 2 | Janet Leverling  | Sales Representative     | 213,051.3                     | 127                    | 321             | 663.71                 | 1,677.57             | 10,238.46                 | 202,812.84                | 4.81                          |
| 3 | Nancy Davolio    | Sales Representative     | 202,143.71                    | 123                    | 345             | 585.92                 | 1,643.44             | 10,036.11                 | 192,107.6                 | 4.96                          |
| 4 | Andrew Fuller    | Vice President, Sales    | 177,749.26                    | 96                     | 241             | 737.55                 | 1,851.55             | 11,211.51                 | 166,537.76                | 6.31                          |
| 5 | Laura Callahan   | Inside Sales Coordinator | 133,301.03                    | 104                    | 260             | 512.7                  | 1,281.74             | 6,438.75                  | 126,862.28                | 4.83                          |
| 6 | Robert King      | Sales Representative     | 141,295.99                    | 72                     | 176             | 802.82                 | 1,962.44             | 16,727.76                 | 124,568.23                | 11.84                         |
| 7 | Anne Dodsworth   | Sales Representative     | 82,964                        | 43                     | 107             | 775.36                 | 1,929.4              | 5,655.93                  | 77,308.07                 | 6.82                          |
| 8 | Michael Suyama   | Sales Representative     | 78,198.1                      | 67                     | 168             | 465.46                 | 1,167.14             | 4,284.97                  | 73,913.13                 | 5.48                          |
| 9 | Steven Buchanan  | Sales Manager            | 75,567.75                     | 42                     | 117             | 645.88                 | 1,799.23             | 6,775.47                  | 68,792.28                 | 8.97                          |

### c. Key insights and Findings

This table shows the sales performance for each employee, with a set of comprehensive indicators from multiple perspectives, enabling the Sales Team to evaluate both revenue generation and pricing negotiation capabilities.



## 13. Business Question 10

### a. Description

This evaluation, requested by the Sales Team, analyzes each employee's performance in their categories, showing the category name, their full name and total sales with discount. It also covers the percentage of each employee's sales within all categories and across all employees.

### b. Results

It is clear that in Beverages category, Margaret Peacock recorded the highest sales amount, reaching \$50,308.21, while Michael Suyama performed the lowest across all metrics, with merely \$9,450.2. In terms of Sales\_staff\_ratio, the highest percentage was 0.25408, achieved by Anne Dodsworth.

|    | AZ category_name | AZ full_name     | 123 total_sales_amount | 123 salse_staff_ratio | 123 total_amount |
|----|------------------|------------------|------------------------|-----------------------|------------------|
| 1  | Beverages        | Margaret Peacock | 50,308.21              | 0.21602               | 0.03974          |
| 2  | Beverages        | Nancy Davolio    | 46,599.36              | 0.24257               | 0.03681          |
| 3  | Beverages        | Janet Leverling  | 44,757.41              | 0.22068               | 0.03536          |
| 4  | Beverages        | Andrew Fuller    | 40,248.25              | 0.24168               | 0.0318           |
| 5  | Beverages        | Robert King      | 27,963.83              | 0.22449               | 0.02209          |
| 6  | Beverages        | Anne Dodsworth   | 19,642.56              | 0.25408               | 0.01552          |
| 7  | Beverages        | Laura Callahan   | 17,897.85              | 0.14108               | 0.01414          |
| 8  | Beverages        | Steven Buchanan  | 11,000.53              | 0.15991               | 0.00869          |
| 9  | Beverages        | Michael Suyama   | 9,450.2                | 0.12786               | 0.00747          |
| 10 | Condiments       | Margaret Peacock | 23,314.87              | 0.10011               | 0.01842          |
| 11 | Condiments       | Andrew Fuller    | 14,850.67              | 0.08917               | 0.01173          |
| 12 | Condiments       | Laura Callahan   | 14,637.66              | 0.11538               | 0.01156          |
| 13 | Condiments       | Nancy Davolio    | 13,561.56              | 0.07059               | 0.01071          |
| 14 | Condiments       | Janet Leverling  | 13,381.64              | 0.06598               | 0.01057          |
| 15 | Condiments       | Anne Dodsworth   | 10,125.55              | 0.13098               | 0.008            |
| 16 | Condiments       | Robert King      | 8,851.38               | 0.07106               | 0.00699          |
| 17 | Condiments       | Michael Suyama   | 4,648.47               | 0.06289               | 0.00367          |
| 18 | Condiments       | Steven Buchanan  | 2,675.3                | 0.03889               | 0.00211          |
| 19 | Confections      | Janet Leverling  | 33,622.4               | 0.16578               | 0.02656          |
| 20 | Confections      | Nancy Davolio    | 28,568.92              | 0.14871               | 0.02257          |
| 21 | Confections      | Margaret Peacock | 27,768.73              | 0.11923               | 0.02194          |
| 22 | Confections      | Laura Callahan   | 21,699.91              | 0.17105               | 0.01714          |
| 23 | Confections      | Andrew Fuller    | 21,455.69              | 0.12883               | 0.01695          |
| 24 | Confections      | Robert King      | 14,518.99              | 0.11655               | 0.01147          |
| 25 | Confections      | Anne Dodsworth   | 8,053.16               | 0.10417               | 0.00636          |
| 26 | Confections      | Michael Suyama   | 6,859.63               | 0.09281               | 0.00542          |
| 27 | Confections      | Steven Buchanan  | 4,809.8                | 0.06992               | 0.0038           |
| 28 | Dairy Products   | Nancy Davolio    | 36,022.98              | 0.18751               | 0.02846          |
| 29 | Dairy Products   | Margaret Peacock | 33,549.8               | 0.14406               | 0.0265           |
| 30 | Dairy Products   | Janet Leverling  | 32,320.84              | 0.15936               | 0.02553          |
| 31 | Dairy Products   | Robert King      | 27,621.86              | 0.22174               | 0.02182          |
| 32 | Dairy Products   | Andrew Fuller    | 23,812.55              | 0.14299               | 0.01881          |
| 33 | Dairy Products   | Steven Buchanan  | 21,937.63              | 0.3189                | 0.01733          |
| 34 | Dairy Products   | Laura Callahan   | 21,101.47              | 0.16633               | 0.01667          |
| 35 | Dairy Products   | Anne Dodsworth   | 21,101.13              | 0.27295               | 0.01667          |
| 36 | Dairy Products   | Michael Suyama   | 17,039.04              | 0.23053               | 0.01346          |

### c. Key insights and Findings

This data demonstrates how the sales performance for each employee in every category, which gives the company an efficient comparison to identify their sales capabilities across different categories.





## 14. Conclusion

- This report delivers a thorough SQL-based analysis tackling main business questions across sales performance, pricing, and category-level contributions. This also successfully builds multiple KPI and provides useful queries for product selecting, range of sales, calculation of percentage.
- These findings reveal that the most contribution is from few specific high-performance employees, which means this company should focus on human resource development to maintain the revenue of this company.
- Each query from several teams is achieved, including selecting specific pricing product for the Product Team, KPI for the Sales Team, human resource structure for the HR Team.



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Instructions: Include a list of references used throughout the project report, following the appropriate citation style.

